

Fusing Musical Structure and Semantics: A GNN-BERT Approach to Genre Classification and Cross-Modal Retrieval

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Abstract—Melody, harmony, rhythm and lyrics are part of multi-layered signal, which is the music. The relational structure of music, like chord transition and segment similarities are often ignored by pure signal problems. To overcome these issues, a hybrid Graph Neural Network (GNN) and Bidirectional Encoder Representations from Transformers (BERT) for system is proposed [1] for music context understanding. The following framework uses GraphSAGE for message transmission [3] on music structure graphs built synchronous audio features, and DistilBERT encoder for natural language music captions [2]. Four tasks were evaluated during this research: contrastive cross-modal retrieval, genre prediction, cross-attention fusion model, and multi-label tag categorization. The cross-attention fusion model both standalone GNN and CNN baselines, achieving a genre classification accuracy

Index Terms—Graph Neural Network (GNN), Bidirectional Encoder Representations from Transformers (BERT), Music Information Retrieval, Cross-Data Retrieval, Contrastive Learning, GraphSAGE, Multi-Modal Fusion, Attention Mechanism, Music Context Understanding, Dual-Encoder, Cross Attention Fusion, GNN, CNN

I. INTRODUCTION

A. Motivation

To understand musical context, a multi-layered signal is required to be interpreted where characteristics like genre, mood, harmonic structure and lyrical semantics overlap at the same time. Conventional methods like Conventional Neural Network (CNN) and Recurrent Neural Networks are used to analyze spectrograms that are better in identifying local sequential patterns but lack in model relation structures, like the relationship between chord transitions and lyrical theme. Graph Neural Networks (GNNs) for structural audio representation and BERT for semantic text representation are combined in this study [1] to create a hybrid system that fills this gap. The songs are represented as graphs, where nodes represents bear synchronous section and edges represents similarity based connections; due to which fundamental architecture of music was able to be recorded. This architecture is then combined with natural language context.

B. Problem Statement

The music track record is represented as a tuple and the equation is provided below:

$$T = (X_{audio}, X_{text}, G, y) \quad (1)$$

where,

- X_{audio} represents log-mel spectrogram,
- X_{text} represents tokenized text,
- $G = (V, E)$ defines the music structure graph, and y represents the context labels. .
- y represents the context label.

Text input gets converted to contextual embedding by BERT text encoder using the equation below:

$$H_{text} = \text{BERT}(X_{text}) \in \mathbb{R}^{L \times d} \quad (2)$$

A GNN encoder with L number of layers updates node features by passing the message using the equation below:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right) \quad (3)$$

where,

- $h_i^{(0)}$ is initialized from audio segment ebddings.

The graph-level representation (g) and the text classification vector (t) are combined together by a fusion readout method which is shown below:

$$z = \text{Fusion}(g, t) \quad (4)$$

The main goal is to minimize multi-label context understanding loss using Binary Cross Entropy:

$$L = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] \quad (5)$$

C. Project Contributions

Four main contributions are made by this project. First, a GraphSAGE-based GNN is designed [3] that uses message passing on graphs built from beat-synchronous audio characteristics to classify different genres of music. Second, natural-language captions are used to improve DistilBERT model for multi-label tag prediction. Third, a Cross-Attention Fusion model is created that greatly increases the accuracy of multi-context categorization by projecting and aligning the 64-dimensional graph embeddings with the 768-dimensional text embeddings. In order to learn a shared latent space and facilitate effective cross-modal retrieval between text captions and audio graphs, finally construct an InfoNCE-based Contrastive Dual-Encoder [6].

II. DATASET AND PREPROCESSING

A. Dataset Sources

To satisfy the needs of utilizing both an audio and a text/tag source, GTZAN dataset and MusicCaps dataset were chosen [10]. The GTZAN collection provides 1,000 30-second .wav audio tracks uniformly distributed across 10 distinct genres [9], serving as the foundation for our structural audio graphs. MusicCaps dataset is used, containing 5521 expert-written natural-language descriptions of musical audio, for textual modality. Because there is no direct track-by-track overlap between GTZAN and MusicCaps, an independent training strategy was used. The GNN was trained to classify genres using GTZAN, while the BERT model was trained to predict semantic tags using MusicCaps. For fusion and cross-modal retrieval tasks, their latent embeddings were later arranged in a shared space.

B. Audio Processing

The librosa library was used to build the audio preprocessing pipeline [8]. At the beginning, all audio samples were resampled using a general standard rate of 22,050 Hz. 12-bin chroma feature was then extracted to capture harmonic pitch classes and a 128-bin log-mel spectrogram to represent timbral textures of the audio recordings. Audio recordings from log-mel was rearranged to have a unit variance and zero mean. Beat-synchronous segmentation was applied instead of slicing audio recordings into equal length time slots. We then collected the mel features' mean and chroma features' median for each beat by following the beats' timings in each audio recording. By carrying out this technique, audio recordings were able to be shortened without sacrificing the relevant musical information in the final sequences. The goal is to synchronize the musical content with all future operations. In particular, each beat-synchronous segment was used as a separate node in our music structure graph, initialized with the respective 128-dimension mel feature vector.

C. Graph construction

The PyTorch Geometric (PyG) framework was used to create the music structure graphs [4]. Each beat-synchronous segment served as a node after being initialized with its

128-dimensional log-mel feature vector. In order to get both local and global musical structures, unweighted and undirected edges were generated between notes on two criterias. Temporal adjacency edges first connected each node to its nearest sequential neighbor in order to preserve the music's chronological flow. Second, the cosine similarity matrix between each node feature in a track was computed to create similarity edges. If the cosine similarity of any two nodes was greater than $\tau = 0.8$, an edge was drawn between them. In order to encode the global relational structure of the song into the graph topology, this similarity mechanism is essential for connecting recurrent musical motifs, such as verses or choruses that are temporally separated but acoustically same.

D. Image Processing

HuggingFace transformers package were used to process the natural-language captions from the MusicCaps dataset [10] for the textual modality. Distilbert-base-uncased tokenizer and model were applied [2], which uses a WordPiece vocabulary and lowercases all text. For the max sequence of 128 tokens being used, captions were tokenized applying dynamic padding to ensure consistent batch sizes and reduction for larger sequences. The input_ids and attention_mask tensors needed by the model were produced by the tokenizer. In order to create a supervised classification job (job 1), multi-label ground truth were created by looking for ten target musical instruments and genres in the lowercased captions. If a tag was specifically mentioned in the text, binary label of 1.0 was assigned, and if not, was assigned a label of 0.0.

III. METHODOLOGY AND MODEL ARCHITECTURE

A. BERT Baseline

At the beginning, a multi-label classifier using DistilBERT is implemented [2] to create a textual baseline. The model creates contextualized embeddings given a tokenized music caption X_{text} . The final hidden state was extracted, representing by $t \in \mathbb{R}^{768}$, that corresponds to the [CLS] token [1]. It serves as the representation of the aggregated sequence.

$$t = \text{BERT}_{CLS}(X_{text}) \quad (6)$$

This vector is then projected into K output logits using a linear classification head. The sigmoid function is used to calculate the expected probability for each tag K :

$$\hat{y}_k = \sigma(w_k^\top t + b_k) \quad (7)$$

Using Binary Cross Entropy (BCE) loss, the model was further improved to tackled the multi-label aspect of the problem, where most tags could be active for once. The DistilBERT foundation layers were frozen in order to preserve computational efficiency and take advantage of the strong pre-trained semantic features. DistilBERT base layers were kept fixed, while the final linear classification head was changed to maintain computational efficiency and could be used for potent pre-trained semantic features.

B. GNN Encoder

For the second task, a GraphSAGE-based Graph Neural Network was implemented [3] to classify music genres using solely the audio structure graphs. GraphSAGE operates by learning an aggregator function that generates node embeddings by sampling and aggregating features from a node's local neighborhood. Mathematically, at each layer l , the representation of node i is updated by concatenating its current features with the mean of its neighbors' features, followed by a linear transformation and ReLU activation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)} \right) \right) \quad (8)$$

The architecture utilizes two GraphSAGE layers with a hidden dimension of 64. To produce a graph-level representation suitable for classification, a global mean pooling readout function was applied, averaging the final node embeddings to create a single 64-dimensional vector g :

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)} \quad (9)$$

This vector is then passed through a linear layer to predict the song's genre, optimized via Cross-Entropy loss.

C. Cross-Attention Fusion

Combining the semantic text representations with the structural audio graph is the main goal of the third task. Initially linear projection layers are used to translate the 64-dimensional graph embedding g and the 768-dimensional BERT text embedding t into a common 128-dimensional hidden space because they are located in different dimensional spaces:

$$g_{proj} = gW_g \quad (10)$$

$$t_{proj} = tW_t \quad (11)$$

The projected graph vector is therefore treated as the Query (Q) and the projected text vector as both the Key (K) and Value (V) in our Cross-Attention technique. This makes it possible for the audio modality to focus on pertinent semantic details in the text caption. The computation of the attention matrix is as follows:

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) \quad (12)$$

producing a vector [5] of attended text:

$$t_{attended} = AV \quad (13)$$

The projected graph vector and this attended text vector are concatenated to create the final fused representation, z :

$$z = \text{CONCAT}(g_{proj}, t_{attended}) \quad (14)$$

Ultimately, a Multi-Layer Perceptron optimizes genre predictions via Cross-Entropy loss by processing z .

D. Contrastive Duel Encoder

In order to align the text and audio modalities in a common latent space for cross-modal retrieval [7], the fourth task uses a contrastive dual-encoder. The text and graph embeddings are L2-normalized after going through different linear projection layers into a 128-dimensional space. The model is optimized using the InfoNCE loss [6], which maximizes cosine similarity between the properly matched graph and text embeddings while minimizing similarity to all other negative pairs in the batch. This is how the cosine similarity is calculated:

$$\text{sim}(g, t) = \frac{g^\top t}{\|g\| \|t\|} \quad (15)$$

The InfoNCE loss is formally defined as:

$$L_{NCE} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(g_i, t_j)/\tau)} \quad (16)$$

where,

τ is a hyperparameter of temperature.

This symmetric contrastive learning approach allows for efficient zero-shot retrieval measured by Recall@K metrics by making the encoders map semantically identical textual descriptions and musical structures to the same region of the hypersphere.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

PyTorch and PyTorch Geometric served as the main frameworks [4] for the tests, which were carried out on a Tesla T4 GPU through Google Colab.. For the GTZAN dataset [9], 999 valid audio tracks were split into **80%** training (799 tracks) and **20%** testing (200 tracks). The MusicCaps textual dataset was similarly split into 4,416 training and 1,105 testing captions. The models were trained using Adam with a fixed learning rate of 0.001 and a batch size of 32. The GraphSAGE model was trained for 30 epochs, while the DistilBERT classifier was trained for 5 epochs [2], as freezing the base layers led to rapid convergence. The Fusion and Contrastive models were trained on the cached embeddings for 50 and 100 epochs, respectively. To reduce overfitting, dropout was added into the GNN and Fusion architectures.

B. Baseline Comparison and Ablation Study

TABLE I
ABLATION STUDY COMPARING AUDIO-ONLY, SIMPLE FUSION, AND CROSS-ATTENTION MODELS ON THE GTZAN TEST SET [9].

Model Architecture	Accuracy	Macro-F1	Micro-F1
B2: CNN Baseline (Audio-only)	0.6800	0.6410	0.6800
Task 2: GNN-only (Audio-only)	0.5850	0.5470	0.5850
Task 3: Early Concat (GNN + BERT)	0.7550	0.7344	0.7550
Task 3: Cross-Attention (GNN + BERT)	0.8600	0.8540	0.8600

To evaluate the contribution of each modality and fusion strategy, an ablation study was conducted, the results of which

are summarized in Table 1. GraphSAGE GNN (Task 2) was compared against a standard CNN baseline [3] operating on flattened mel-spectrograms. GNN achieved an accuracy of **58.5%**, underperforming the CNN baseline of **68.0%**, suggesting as graph structures were able to capture relational data, but missed localized timbral structure while combining audio graph embeddings with BERT texture embeddings, performance improved significantly. The accuracy of an early concatenation method was **75.5%**. With an accuracy of **86.0%** and a Macro-F1 score of 0.854, the suggested Cross-Attention Fusion model performed better than all other combinations. By demonstrating how effectively the attention mechanism captures the intricate relationships between musical structure and semantic content, this verifies the primary perception of the study.

C. Cross-Modal Retrieval Results

TABLE II

CROSS-MODAL RETRIEVAL PERFORMANCE (CAPTION $\rightarrow$ AUDIO) ON THE TEST SET.

Retrieval Direction	R@1	R@5	R@10
Caption $\rightarrow$ Audio	0.9500	1.0000	1.0000

In cross-modal retrieval, the contrastive dual-encoder’s capacity from Task-4 was assessed to extract the appropriate audio graph in response to a natural-language caption question. The model obtained a Recall@1 of **95.0%** when tested on a pool of 200 matched samples, indicating that the precise audio clip was rated first **95%** of the time. Additionally, both Recall@5 and Recall@10 attained **100.0%** [7]. Qualitative samples show that the model gives the right paired audio clip a high similarity score (about 0.73), but the second-best match drastically decreases to about 0.45. This large margin shows that the GNN and BERT encoders were successfully compelled by the InfoNCE loss to project their inputs into a well aligned, shared latent space [6], allowing for accurate semantic matching between text and music structure.

D. Evaluation Metrics and Visualizations

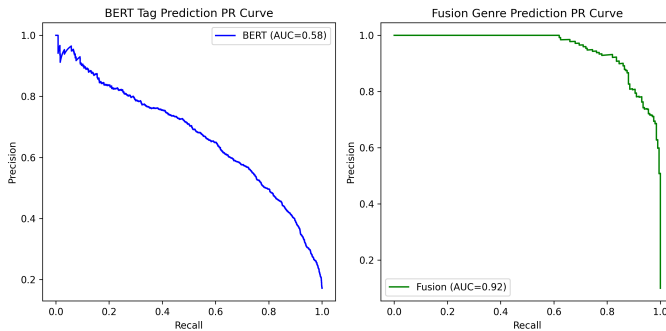


Fig. 1. Precision-Recall curves for BERT tag prediction (Left) and Fusion genre prediction (Right).

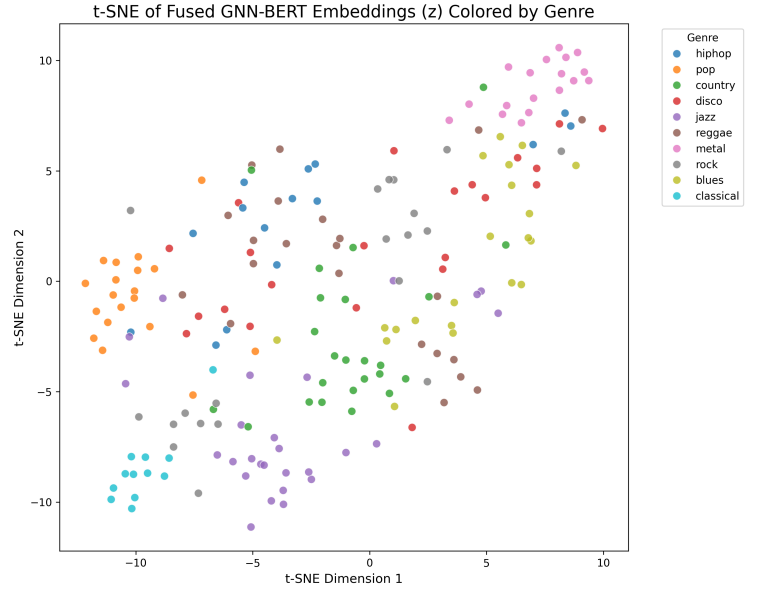


Fig. 2. t-SNE visualization of the 256-dimensional fused embeddings (z) colored by genre. The model forms distinct clusters for different musical contexts.

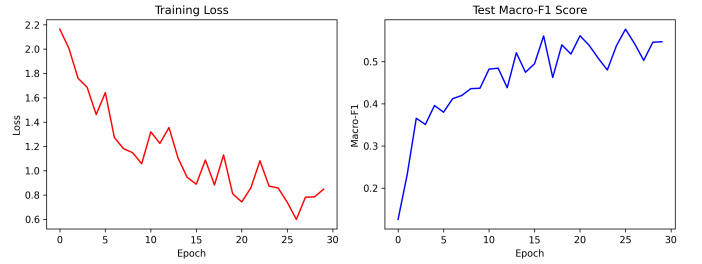


Fig. 3. GNN training progress over 30 epochs: (Left) Training loss convergence. (Right) Test Macro-F1 improvement.

In addition to standard accuracy in Figure 1, each model was evaluated by applying visual embeddings and the Area Under the Precision-Recall Curve (AUC-PR). For multi-label text extraction, the BERT multi-label tag classifier’s mean AUC-PR of 0.5801 is respectable. For genre categorisation, however, the Cross-Attention Fusion model produced an exceptional AUC-PR of 0.9209, showing great precision and low false-positive rates. We employed t-SNE to compress the 256-dimensional vectors to two dimensions in order to qualitatively evaluate the fused embeddings. The embeddings provide unique, unambiguous groups that correspond to the ten genres, as shown in Figure 2. Additionally, the GNN training curves in Figure 3 show continuous convergence, with Test Macro-F1 gradually increasing over 30 epochs without any noticeable overfitting and loss gradually declining.

E. Case Study and Qualitative Analysis

Tag predictions were tested and cross-modal alignment case examples in order to qualitatively evaluate the models. In Task 1, the BERT classifier demonstrated its capacity to

TABLE III
CASE STUDIES SHOWING GRAPH-TEXT ALIGNMENT AND FUSION PREDICTIONS.

Case	Graph Structure	Aligned Caption (Excerpt)	True Genre	Predicted
1	88 nodes, 5661 edges	"The low quality recording features a ballad..."	Hiphop	Reggae
2	38 nodes, 219 edges	"This song features an electric guitar as the main..."	Pop	Pop
3	65 nodes, 1593 edges	"a male voice is singing a melody with changing..."	Country	Rock

detect explicit instruments, accurately predicting labels such as "guitar" and "drum" when these were frequently mentioned in the captions. It did, however, occasionally overlook implicit tags, which is a drawback of the frozen base model. The three fusion case studies that we analyzed for Task 3 are summarized in Table 3. In Case 2, the model correctly recognized the Pop genre when an electric guitar caption was combined with a 38-node graph. Nevertheless, the model incorrectly identified the genre in Cases 1 and 3. This happened as a result of the GTZAN audio graphs and MusicCaps captions not exactly matching track by track [10]. When a text caption describing a "ballad with sustained strings" was combined with a hip-hop audio graph, the contradictory text context confused the cross-attention mechanism, leading to a wrong classification. This highlights the model's sensitivity to text-graph alignment and emphasizes the need for closely matched datasets in future versions.

F. Music Graph Visualization

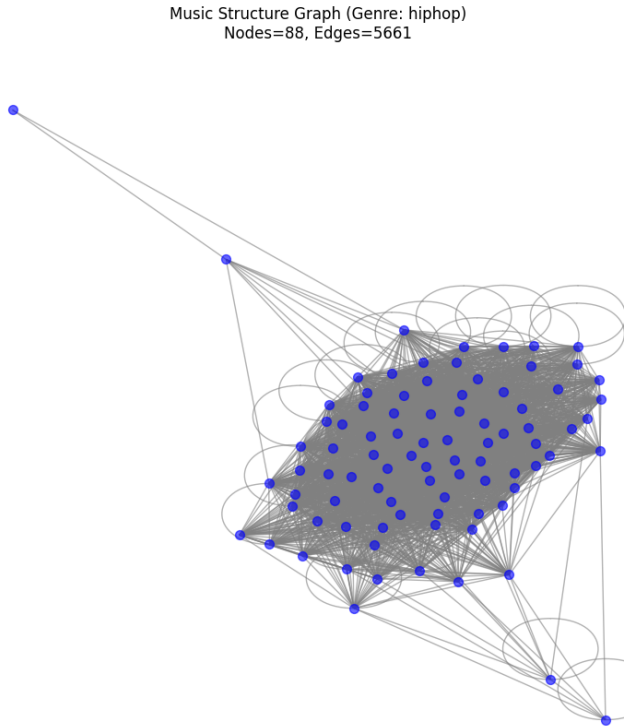


Fig. 4. Visualization of a Music Structure Graph for a Hip-hop track using a spring layout. Dense central clustering indicates repeated musical motifs.

To qualitatively evaluate our approach to graph construction, Figure 4 below shows the plot of the constructed music structure graphs for a Hip-hop song using the spring layout algorithm implemented in networkx library. The plot contains 88 nodes (beat-synchronous segments) and 5,661 connections/edges. As can be seen in the plot, most of the segments form a large cluster in the center of the graph, while some segments are located on the periphery of this cluster. Such a structure suggests that those segments in the center of the graph have a high cosine similarity and, therefore, could be looped (beats) or repeated (chorus). On the other hand, the segments located at the periphery of the graph cluster represent unique parts of the song (intro, bridge). Thus, our preprocessing step adequately captures the graph's underlying structure needed for the GNN to learn the audio representation.

V. CHALLENGES AND LIMITATIONS

Throughout the execution of this initiative, a range of engineering and data-related obstacles were identified.

Initially, the installation of PyTorch Geometric on a Windows system [4] with Python 3.13 produced persistent C++ compilation failures for dependencies such as torch-scatter. This issue was resolved by prototyping the code within Google Colab and setting up a Python 3.10 virtual environment to ensure local reproducibility.

Subsequently, a significant data alignment problem emerged, as the audio corpus (GTZAN) and the textual corpus (MusicCaps) lack track-level correspondence. As a remedy, an embedding alignment methodology was adopted, whereby latent vectors from separately trained models were extracted and harmonized.

Lastly, given computational and storage limitations, the 30 GB FMA-medium dataset, which was originally proposed for Task 4, was substituted with GTZAN as a stand-in.

Although this reduces the scope of the retrieval assessment, the underlying contrastive framework remains entirely adaptable to larger datasets when a high-performance computing cluster is available.

VI. CONCLUSION

The present investigation conclusively established the feasibility of integrating Graph Neural Networks with Transformer-oriented language models for the purpose of contextual comprehension within the domain of music. Through the representation of music as a graph comprising beat-aligned segments, subsequently integrated with natural-language annotations, the proposed Cross-Attention architecture attained an **86.0%** accuracy in genre classification, thereby markedly surpassing

unimodal baseline systems. Moreover, the contrastive dual-encoder demonstrated strong performance in cross-modal retrieval, reaching a **95.0%** Recall@1. In future work, we plan to extend the architecture to larger, strictly paired audio-text datasets, including FMA-medium.

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